10x and cellranger

COURSE ON SCRNA-SEQ DATA ANALYSIS

Fundamentals of 10x Chromium and preprocessing

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with acknowledgments to Deepak Tanwar and SIB

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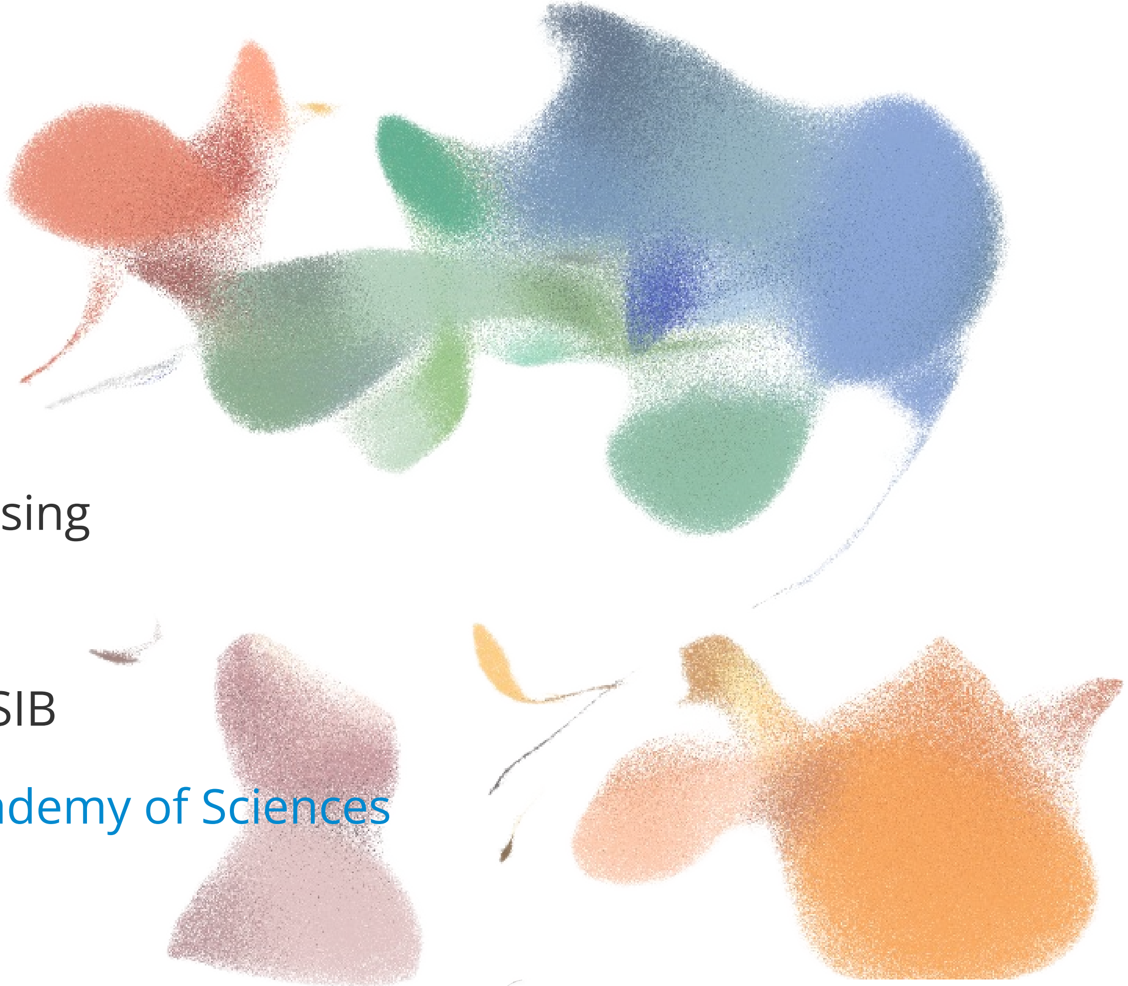
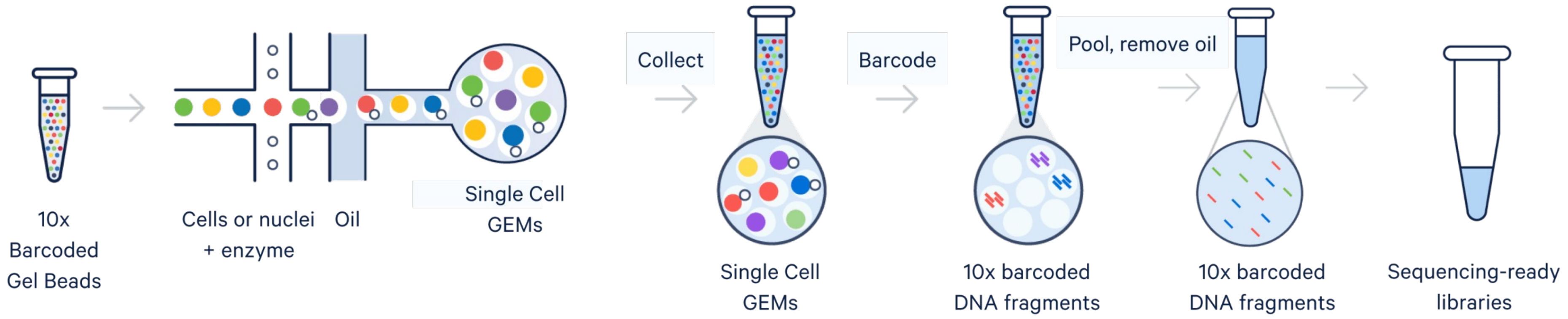


ELIXIR CZECH REPUBLIC

IMG

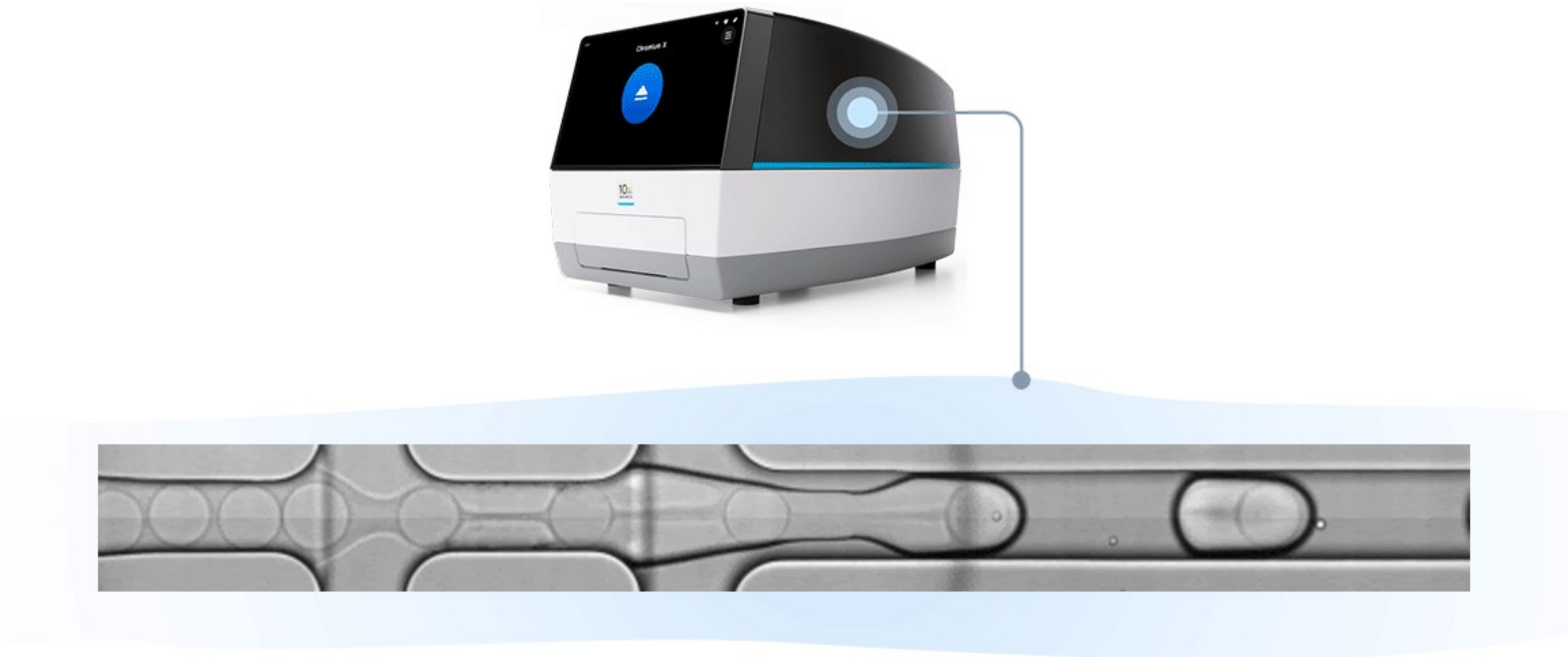
How to capture cell of origin?

- Gel Beads-in-emulsion (GEM)
  - Way to label mRNA from the same cell

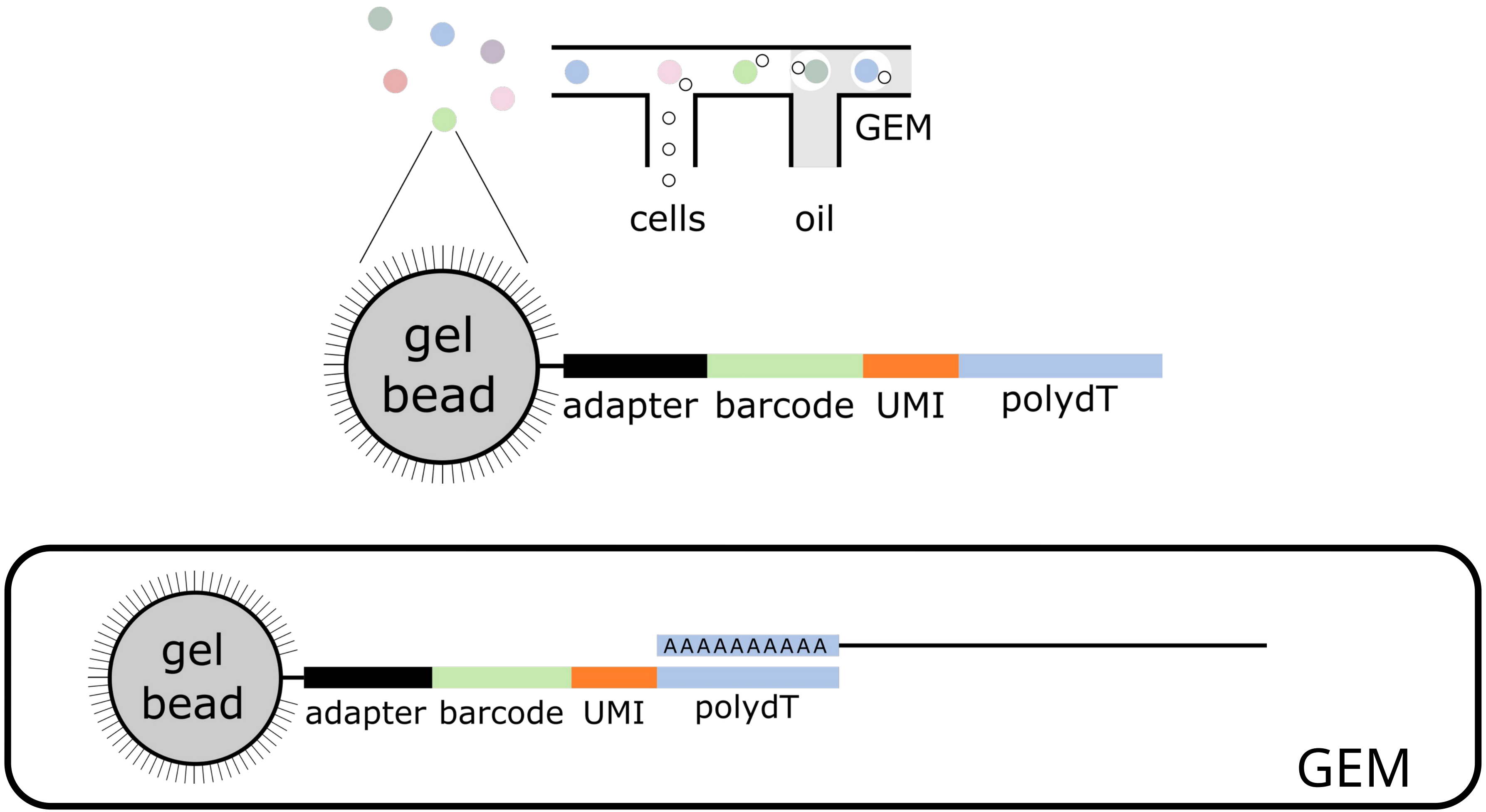


How to deliver the label?

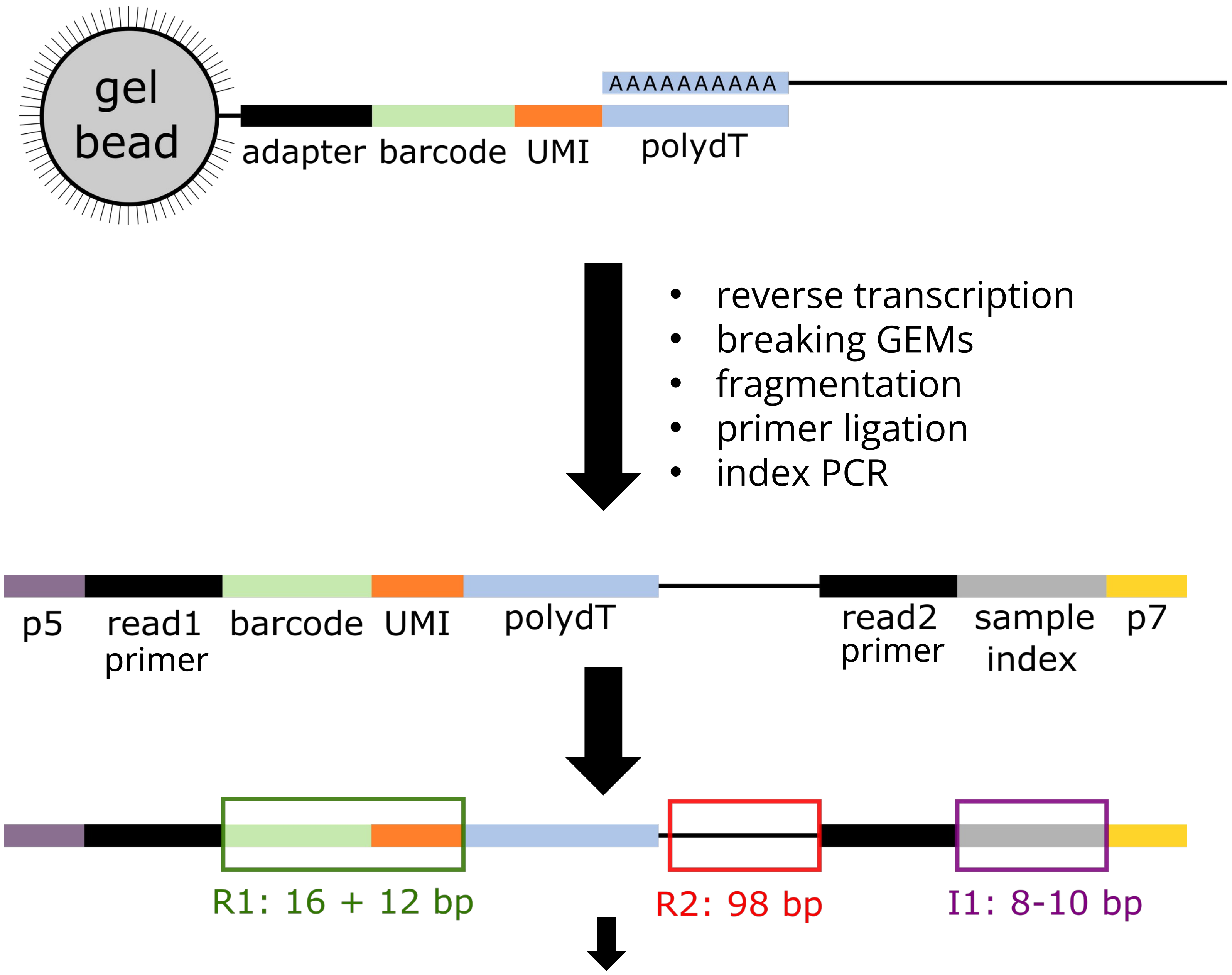
- 10x chromium instrument



10,000+ GEMs in 4 minutes.



All captured transcripts from single cell:
identical + unique barcode



Illumina sequencing

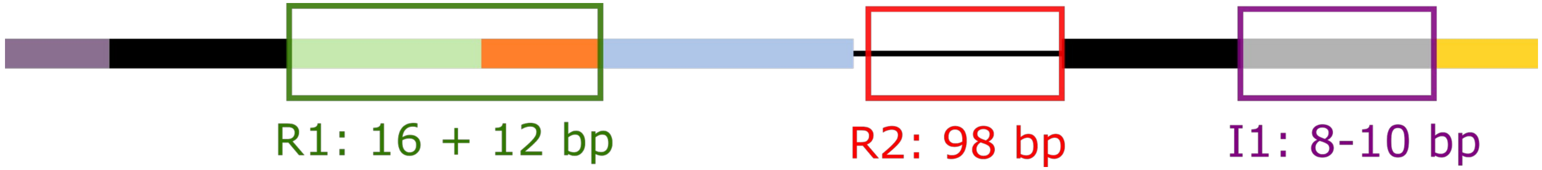
Sequencing output

- Filenames notation

sample id　　　　read type

```
ETV6-RUNX1_1_S1_L001_R1_001.fastq.gz
ETV6-RUNX1_1_S1_L001_R2_001.fastq.gz
ETV6-RUNX1_1_S1_L001_I1_001.fastq.gz
```

lane



- Files from inside

ETV6-RUNX1_1_S1_L001_R1_001.fastq.gz

```
@SRR9264343.187191690 K00271:135:HK3J7BBXX:3:2110:12591:47964 length=26
GGCAATTCATTGGGCCCCCGAAGCTC
+
AAF-FJFJJJJJJJJJJJJJJJJJJJ
@SRR9264343.5519124 K00271:135:HK3J7BBXX:3:1112:2950:8295 length=26
GGCAATTCATTGGGTGAGGTTGTAGG
+
AAFFFJJJJJFFJFJJFJJJJJJJJJ
@SRR9264343.149664941 K00271:135:HK3J7BBXX:3:2217:29579:19267 length=26
TCGGATGTCTGAAAGAGTGCTAGAAT
...
```

ETV6-RUNX1_1_S1_L001_R2_001.fastq.gz

```
@SRR9264343.187191690 K00271:135:HK3J7BBXX:3:2110:12591:47964 length=98
GTAGTAAAAGACTGGTTAATGATAACAATGCATCGTAAAACCTTCAGAAGGAAAGGAGAATGTTTTGTGGACCACTTTCGTTTTCTTTTTTGCGTGTG
+
-777<<A-<FF<<A7-<FAFAFFJJJJJFFFFFFFJFFJJJFFJ7JFFJJJ<FJJAFJJFJJ<A-<AA<FFF)<A--7-7--AFFJ--AFAFA7<AA-
@SRR9264343.5519124 K00271:135:HK3J7BBXX:3:1112:2950:8295 length=98
AGACTTTCTACCTGGTCATATACTCTGCAGCTGTTAGAATGTGCAAGCACTTGGGGACAGCATGAGCTTGCTGTTGTACACAGGGTATTTCTAGAAGC
+
--77-77--7-77<-7AFAJJJJJ7A7AFF7-A-7FFJJFA-AF77FF<-A<FFJJF-)FF7F<7FA7FAAAFJ-A--A<AAFJJ<F----7--<F77
@SRR9264343.149664941 K00271:135:HK3J7BBXX:3:2217:29579:19267 length=98
GGTCCGCAAGCAGGTGGTGCACATCCCGTCCTTCATTGTCCGCCTGGCTTCCCAGAAGCGCATCGACTTCTCTCTGCGCTCTCCCTACGGGGGTGGCC
…
```

ETV6-RUNX1_1_S1_L001_I1_001.fastq.gz

```
@SRR9264343.187191690 K00271:135:HK3J7BBXX:3:2110:12591:47964 length=8
CCTAGACC
+
AAAFFJJJ
@SRR9264343.5519124 K00271:135:HK3J7BBXX:3:1112:2950:8295 length=8
CCTAGACC
+
AAAFFFJF
@SRR9264343.149664941 K00271:135:HK3J7BBXX:3:2217:29579:19267 length=8
CCTAGACC
...
```

Data preprocessing

1. Mapping
2. Quantification
3. Cell calling

- For 10x data use cellranger
- Alternatively Alevin, STARsolo

Cellranger reference data

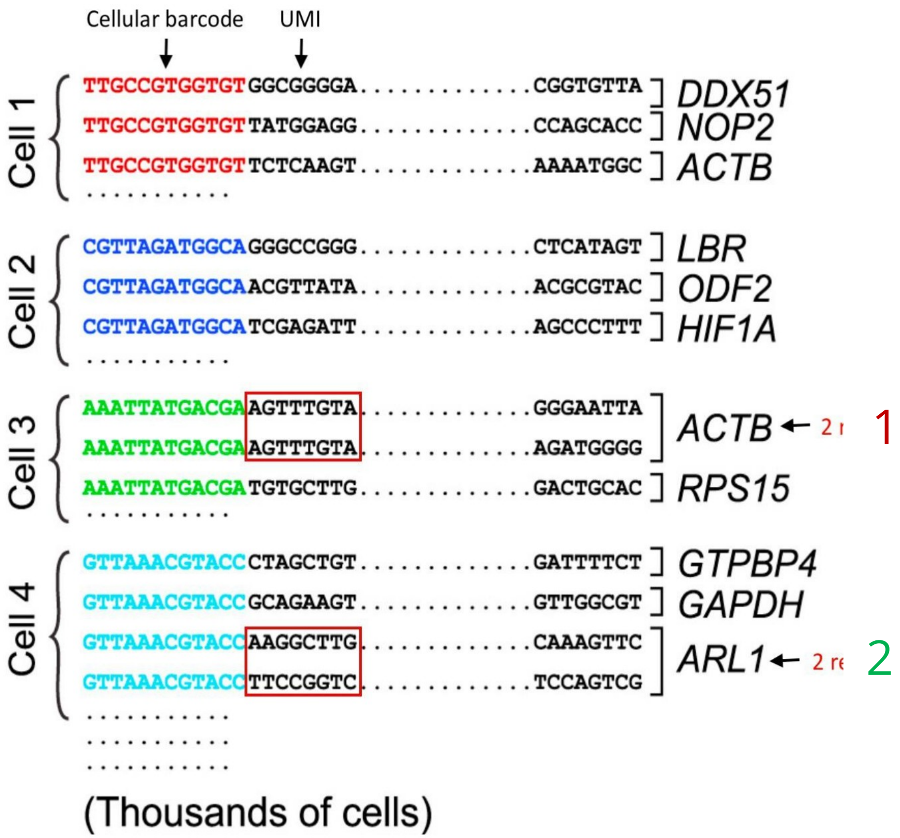
- Human, mouse and rat available for download
- Other organisms can be prepared with `cellranger mkref`
    - Use Ensembl for input data
    - Add exogenous sequences (eGFP etc.)
    - Docs

Counting UMIs vs alignments

- UMI: Unique Molecular Identifier:
  - Identifies each molecule (i.e. sequence) uniquely
- Molecules from a common PCR template -> carry the same UMI
- By counting UMI: correct for PCR replicates

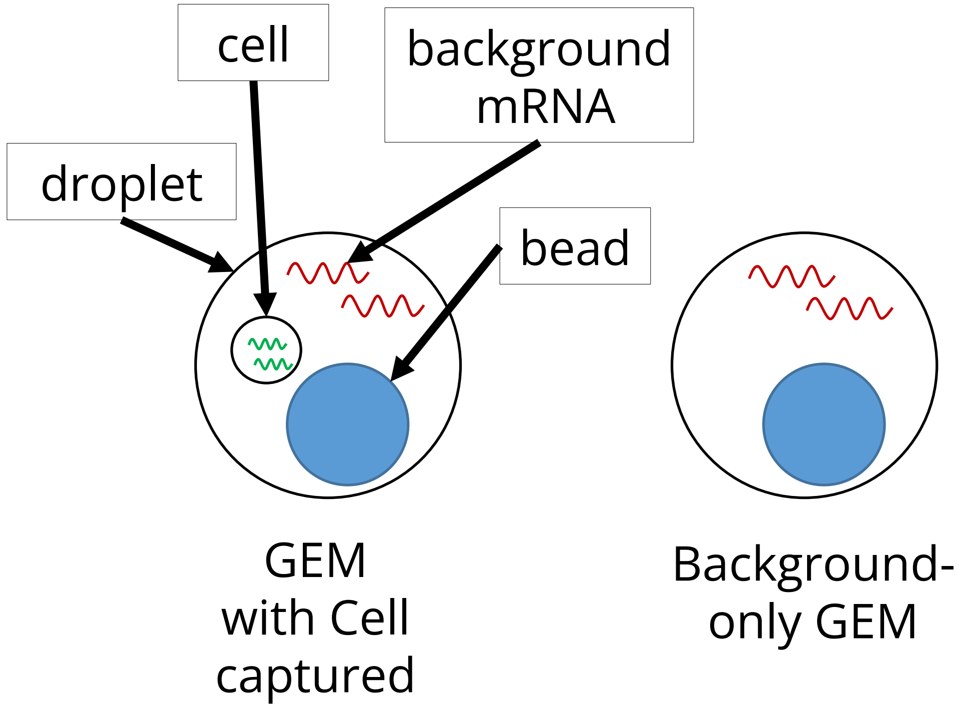
Counting UMIs vs alignments

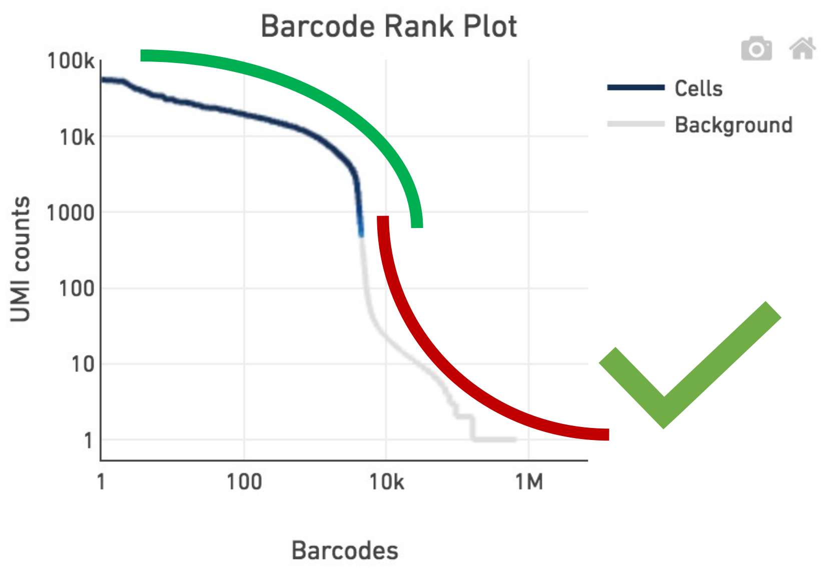
- Reads with different UMIs were derived from different molecules and are biological duplicates
- Reads with the same UMI originated from the same molecule and are technical duplicates – the UMIs should be collapsed to be counted as 1

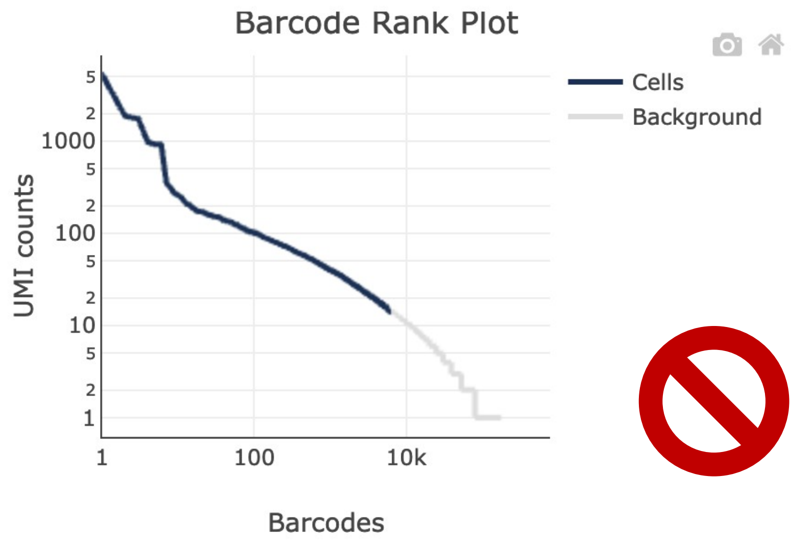


Cell calling

- Cell = high UMI content
- Empty droplet = low UMI content vs backgrounc







Cellranger count output

ETV6-RUNX1_1

Alerts

The analysis detected ⚠ 1 warning.

	Alert	Value	Detail
⚠	Fraction of RNA read bases with Q-score >= 30 is low	59.4%	Fraction of RNA read bases with Q-score >= 30 should be above 65%. A lower fraction might indicate poor sequencing quality. This is Read 1 for the Single Cell 3' v1 chemistry and Single Cell 5' paired end, Read 2 for the Single Cell 3' v2/v3 chemistry and Single Cell 5' R2-only)

Summary | Analysis

3,091
Estimated Number of Cells

68,259
Mean Reads per Cell

1,717
Median Genes per Cell

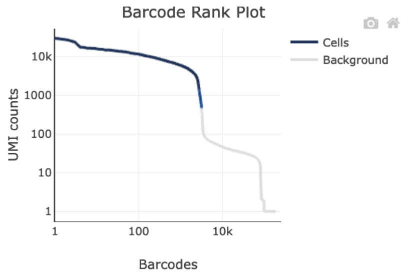
Sequencing

Number of Reads	210,987,037
Number of Short Reads Skipped	0
Valid Barcodes	98.2%
Valid UMIs	100.0%
Sequencing Saturation	84.4%
Q30 Bases in Barcode	96.4%
Q30 Bases in RNA Read	59.4%
Q30 Bases in UMI	96.5%

Mapping

Reads Mapped to Genome	95.8%
Reads Mapped Confidently to Genome	92.9%
Reads Mapped Confidently to Intergenic Regions	5.2%
Reads Mapped Confidently to Intronic Regions	25.5%
Reads Mapped Confidently to Exonic Regions	62.2%
Reads Mapped Confidently to Transcriptome	58.2%
Reads Mapped Antisense to Gene	1.2%

Cells



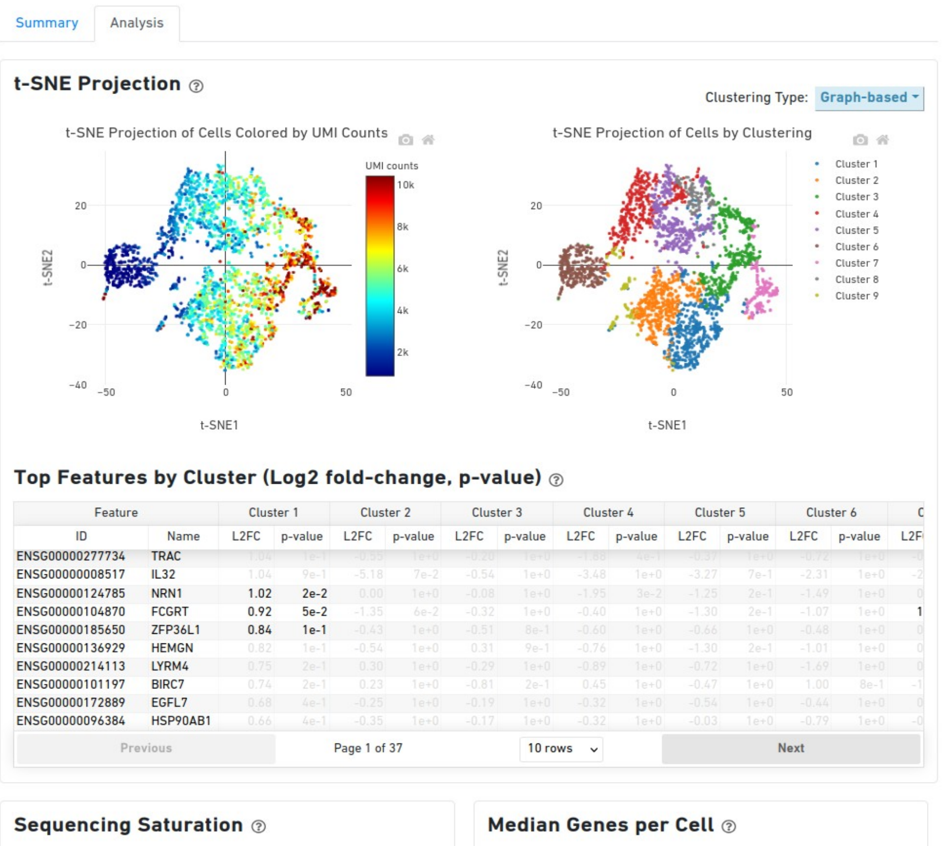
Estimated Number of Cells	3,091
Fraction Reads in Cells	91.1%
Mean Reads per Cell	68,259
Median Genes per Cell	1,717
Total Genes Detected	18,334
Median UMI Counts per Cell	4,825

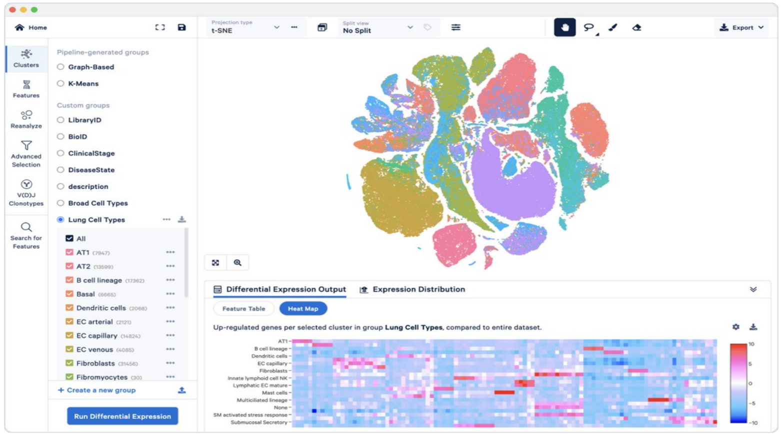
Sample

Sample ID	ETV6-RUNX1_1
Sample Description	
Chemistry	Single Cell 3' v2
Include introns	False
Reference Path	…nger/refdata-cellranger-GRCh38-3.0.0
Transcriptome	GRCh38-3.0.0
Pipeline Version	cellranger-6.0.1

Cellranger count output

- Some tertiary analysis for EDA
- Advanced analysis with 10x Loupe Browser
- Or use third-party tools





10x portfolio for single cell

Flex

Fix, batch, and run on your schedule.

Protein coding gene coverage

Performs well with low-quality and FFPE samples

Multiomic readouts from the same cell

Gene expression
Protein
CRISPR

Throughput options

Up to 8M cells (1–3,072 samples) per run

Products

Flex >

Universal

Species agnostic. Maximum versatility.

Whole transcriptome coverage

Delivers broadest set of information, including isoforms, SNPs, etc.

Multiomic readouts from the same cell

3' or 5' gene expression
TCR/BCR
Protein
CRISPR

Throughput options

Up to 160,000 cells (1–8 samples) per run

Products

Universal 3' >
Universal 5' >

Epi Chromatin

Unmask epigenomic profiles.

ATAC-seq chemistry

Explore open chromatin regions
Link directly to 3' gene expression (Multiome kit)

Multiomic readouts from the same cell

Chromatin accessibility
3' gene expression

Throughput options

Up to 80,000 nuclei (1–8 samples) per run

Products

Epi ATAC >
Epi Multiome >

Thank you for your attention!